

Statquest $\rightarrow$ Josh Starmer

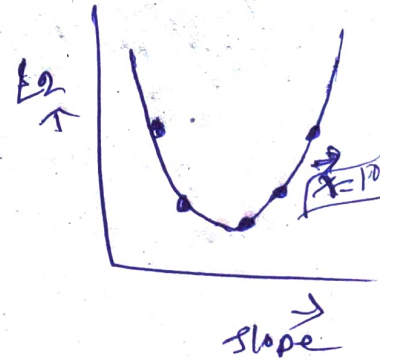
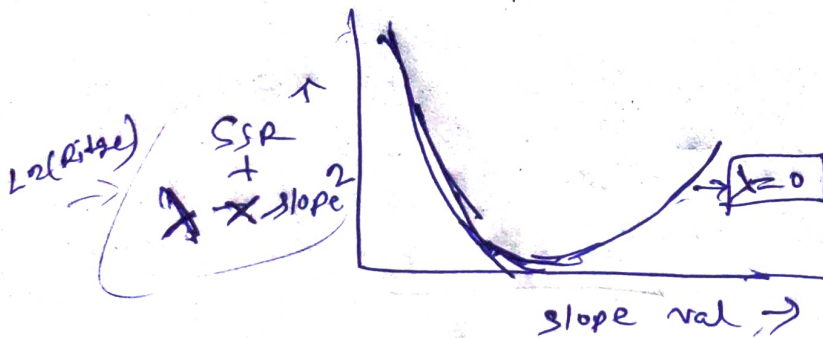
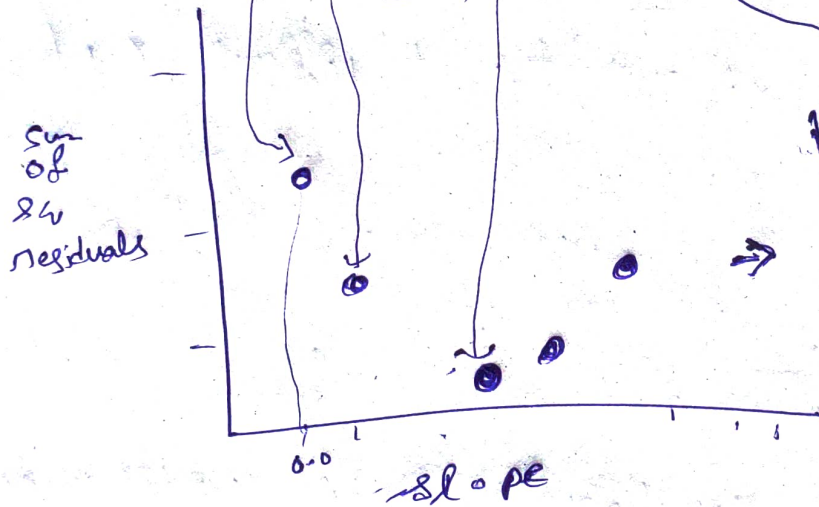
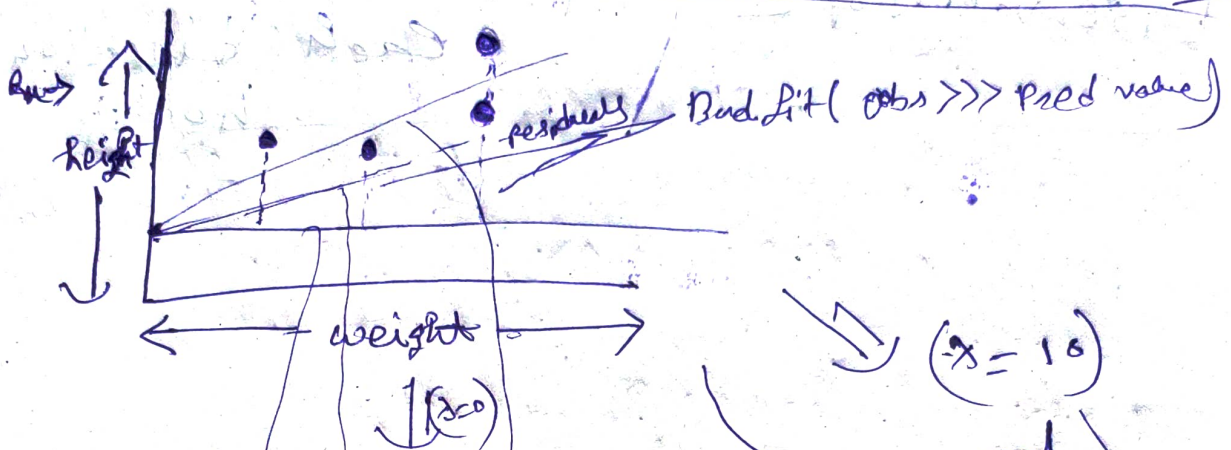
Regularization all $\Rightarrow$

Lect - 28 - 29

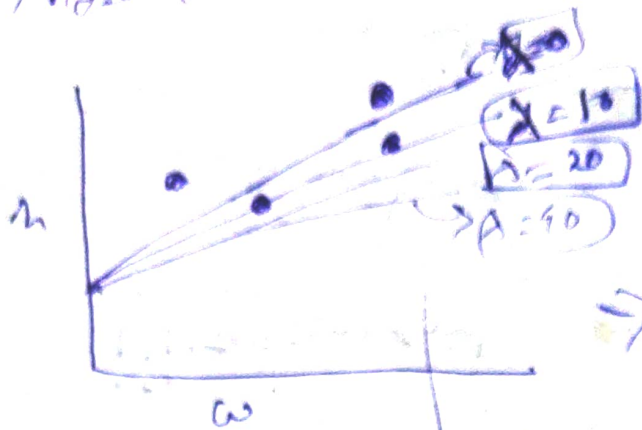
Part 1 (29 Apr)

(Ridge, Lasso, Elastic-net Regression in R)

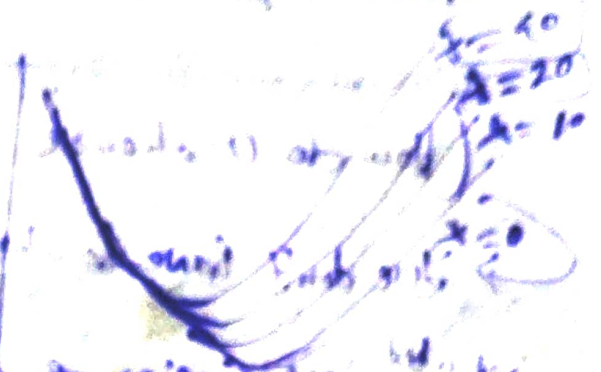
$\Rightarrow$ Ridge Vs Lasso Regression, Visualized



→ After comparison → $\lambda = 10$ ^{penalty} ~~gradient~~ slope.



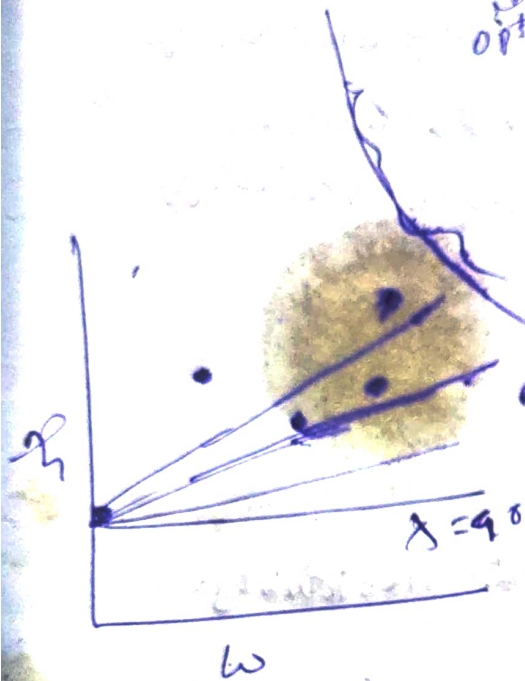
at $\lambda = 10$
 $\rightarrow (1.4)$



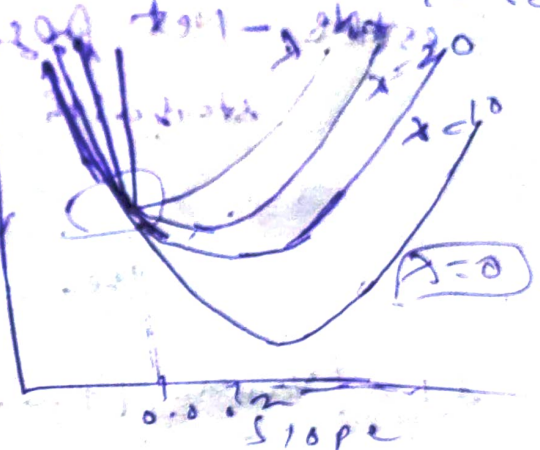
opt slope
 is shrinking
 with + λ

shrink + shrink close
 to zero

but + ... + but doesn't equal
 to zero



$\rightarrow$ L_2
 $\downarrow$
 $(SSR + \lambda |slope|)$



Summary : -

(1) $L_2(\lambda)$ $\Rightarrow$ optimal slope shifts towards
 0, but retain a parabola shape
 $\rightarrow$ never becomes zero.

(2) $L_1(\lambda)$ $\Rightarrow$ optimal shifts towards 0,
 but it ends up to zero,
 '0' becomes optimal slope

19916 Regularization Part 3: Elastic-Net Regression

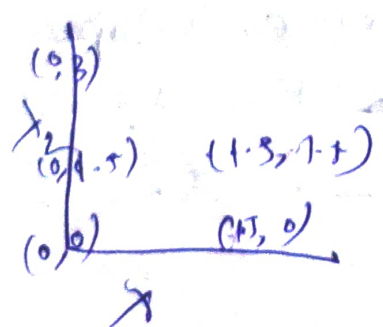
$\lambda_1 = 0$ (Lasso) + Ridge Regression (λ_2)
 if $\lambda_1 = 0$ always by w (Lasso) + Ridge Regression (λ_2)
 if we don't know λ_1 instead use Elastic-Net Regression!!!

Demand of Regression Surfaces is summarized
 size = y intercept + slope $\times$ weight + diet diff
 + slope $\times$ Diet + slope $\times$ Age + slope $\times$ size
 of other variables + ... + long more variables...

Elastic-Net Regression
 starts with least squares
 the sum of squared residuals

$$\lambda_1 \times (|variable_1| + \dots + |variable_x|) + \lambda_2 (variable_1^2 + \dots + variable_x^2)$$
 L1 (Lasso) L2 (Ridge)

we use cross validation of λ_1 & λ_2 to find the best values.



- c-1: $\rightarrow$ when $\lambda_1 = \lambda_2 = 0 \rightarrow SSA + L_1 + L_2$
- c-2: $\rightarrow$ if $\lambda_1 > 0$ & $\lambda_2 = 0 \rightarrow SSA + L_1 + L_2$
- c-3: $\rightarrow$ if $\lambda_1 = 0$ & $\lambda_2 > 0 \rightarrow SSA + L_1 + L_2$

When to use $\rightarrow$

$\Rightarrow$ Regression is good at dealing with situations when there are correlations b/w parameters.

Ridge, Lasso & Elastic-Net Regression

$\rightarrow$ here we use $\rightarrow$ glmnet lib to do Ridge, Lasso, Elastic-Net Regression
(Generalized linear models)

$\rightarrow$ glmnet $\rightarrow$ single λ

the sum of the squared residuals
+

$$\lambda \left[\alpha \times (|\text{variable}_1| + \dots + |\text{variable}_x|) + (1-\alpha) \times (\text{variable}_1^2 + \dots + \text{variable}_x^2) \right]$$

$\alpha = 0 \rightarrow L2(\text{Ridge})$

$\alpha = 1 \rightarrow L1(\text{Lasso})$

λ Controls $\rightarrow$ How much of the penalty to apply

$\lambda = 0 \rightarrow$ no penalty just OLS

when $\lambda = 0$ $\rightarrow$ Elastic-Net penalty comes into the pic

$\rightarrow$ In glmnet $\rightarrow$ use diff values of λ, α

Code $\rightarrow$

$\rightarrow$ library(glmnet)

$\rightarrow$ set.seed(42)

$\rightarrow n \leftarrow 1000$

$\rightarrow p \leftarrow 5000$

$\rightarrow \text{real} \leftarrow p \leftarrow 15$

$\rightarrow x \leftarrow \text{matrix}(\text{rnorm}(n \times p), nrow = n, ncol = p)$

$\rightarrow$ the made up dataset will have $n = 1000$ samples.


```

> alpha0.5.predicted <- predict(alpha0.5.fit,
                                s = alpha0.5.
                                fit.$lambda[10],
                                newx = x.test)

```

4.296073

```

> list.of.fits <- list()

```

```

> for (i in 0:10) {
  fit.name <- paste0("alpha", i/10)

```

```

  list.of.fits[[fit.name]] <-
    @vglmnet(x.train, y.train, type =
      "mse", alpha = i/10,
      family = "gaussian")
}

```

```

> results <- data.frame()

```

```

> for (r in 0:10) {
  fit.name <- paste0("alpha", i/10)

```

```

  Predicted <-
    predict(list.of.fits[[fit.name]],
            s = list.of.fits[[fit.name]]
            $lambda[10], newx = x.test)

```

```

  mse <- mean((y.test - Predicted)^2)

```

```

  temp <- data.frame(alpha = i/10,

```

mse = mse, fit.name

```

  results <- bind(results, temp)
}

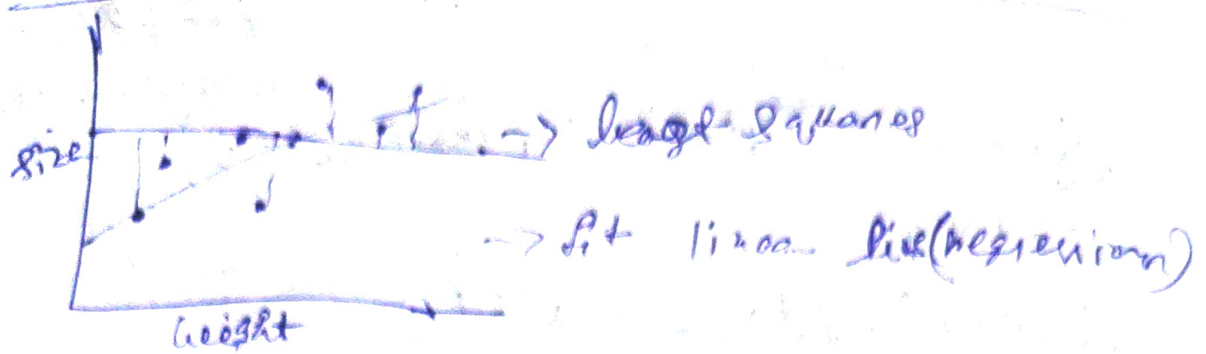
```

```

> results

```

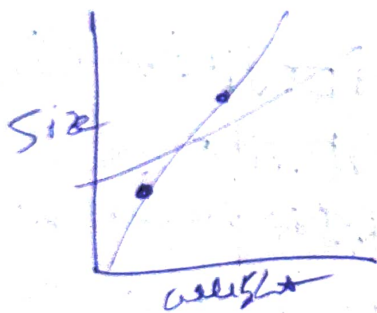
Regularization Part 3: Ridge (L2) Regression



→ problem: — Least squares results in a line that is overfit & has high variance.

↳ Ridge Regression can find a new line that doesn't fit the training data as well... in short we are adding a small bias into how the new line fits to the data...

→ L2 → better for testing line long term predictions.



to $\min(b) \cdot SSR \rightarrow + L2 \text{ penalty}$

$$L2 = SSR + \lambda \times \text{slope}^2$$

Summary: — when the sample size are relatively small ($\downarrow$) then Ridge Regression can improve predictions made from new data (variance) by making the predictions less sensitive to the training data.

→ Done by adding L_2 penalty.

$$\boxed{SSR + \lambda \times \text{slope}^2}$$

determined using Cross validation

→ even when there isn't enough data to find the least squares parameter estimates, L_2 can still find a soln with cross validation & L_2 penalty(λ)

Regularization → L_1 (Lasso)

$$L_{\text{asso}} \rightarrow \boxed{SSR + \lambda \times |\text{slope}|}$$

Ridge v/s Lasso →

→ Ridge can only shrink the slope asymptotically close to 0.

→ That means

it just reduces importance to the useless variables.

→ Lasso Regr... can shrink the slope all way to 0.
That means

→ can include useless variable making final eqn easier & simpler